

Last Time

Review

Sources of Variation

ANOVA Model

- Sum of Squares Within

- Sum of Squares Between

- The F-statistic

Example

- The Summary Table

Effect Size and Power

Today

Review: ANOVA Model

Effect Size and Power

Post-hoc Tests

- Familywise vs. Per Comparison

- Bonferroni Inequality

- Range Tests

GSS 1980

In Class Exercise

The ANOVA Summary Table

Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Between	43.33	2	21.67	4.06
Within	64.00	12	5.33	
Total	107.33	14		

$$F_{.05}(2, 12) = 3.8853$$

$$Pr(F(2, 12) > 4.06 \mid H_0 \text{ true}) = 0.0450$$

The ANOVA Model

Let X_{ij} be the measured variable for Individual i in Group $j = 1, \dots, k$.

	Statistic	Parameter
Group mean	$\bar{X}_j$	μ_j
Grand mean	$\bar{\bar{X}}$	μ
Error	$X_{ij} - \bar{X}_j$	ϵ_{ij}
Effect	$\bar{X}_j - \bar{\bar{X}}$	τ_j

Error $\epsilon_{ij} \sim \mathcal{N}(0, \sigma_e^2)$

The ANOVA Model

Let X_{ij} be the measured variable for Individual i in Group $j = 1, \dots, k$.

► Model:

$$\begin{aligned} X_{ij} &= \mu_j + \epsilon_{ij} \\ &= \mu + (\mu_j - \mu) + \epsilon_{ij} \\ &= \mu + \tau_j + \epsilon_{ij} \end{aligned}$$

The ANOVA Model

Let X_{ij} be the measured variable for Individual i in Group $j = 1, \dots, k$.

$$\begin{aligned} MS_{\text{Within}} &= \frac{1}{N_{\text{Total}} - k} \sum_{j=1}^k \sum_{i=1}^{N_j} (X_{ij} - \bar{X}_j)^2 \\ &= \frac{\sum_{j=1}^k (N_j - 1) s_j^2}{N_{\text{Total}} - k} \end{aligned}$$

$$MS_{\text{Between}} = \frac{1}{k - 1} \sum_{j=1}^k N_j (\bar{X}_j - \bar{\bar{X}})^2$$

$$E(MS_{\text{Within}}) = \sigma_e^2$$

$$E(MS_{\text{Between}}) = \sigma_e^2 + \frac{\sum_{j=1}^k N_j \tau_j^2}{k - 1}$$

The ANOVA Model

Let X_{ij} be the measured variable for Individual i in Group $j = 1, \dots, k$.

$$\begin{aligned} F &\approx \frac{E(MS_{\text{Between}})}{E(MS_{\text{Within}})} \\ &= \frac{\sigma_e^2 + \frac{\sum_{j=1}^k N_j \tau_j^2}{k-1}}{\sigma_e^2} \end{aligned}$$

Assumptions of ANOVA

- ▶ The populations (X_j) are normally distributed
 - ▶ violations of this assumption are not serious if N_{Total} is large

Assumptions of ANOVA

- ▶ The populations (X_j) are normally distributed
 - ▶ violations of this assumption are not serious if N_{Total} is large
- ▶ Homoscedasticity: equal variances across groups
 - ▶ violations are not serious if N_{Total} is large or if N_j s are equal across groups

Assumptions of ANOVA

- ▶ The populations (X_j) are normally distributed
 - ▶ violations of this assumption are not serious if N_{Total} is large
- ▶ Homoscedasticity: equal variances across groups
 - ▶ violations are not serious if N_{Total} is large or if N_j s are equal across groups
- ▶ Each group is independent (otherwise we have a repeated measures design)

ANOVA and Power

The same factors affect power in the ANOVA as in z and t tests:

- ▶ The effect sizes, the true differences between the means
- ▶ The variance of the population (σ_e^2)
- ▶ Numerator $df = k - 1$
- ▶ Denominator $df = N_{\text{Total}} - k$
- ▶ The α -level

ANOVA Effect Size

- ▶ Cohen's d is a distance measure.
- ▶ ANOVA relies on r measures.
- ▶ r measures are a function of the correlation between dependent and independent variables.

ANOVA Effect Size

- ▶ Cohen's d is a distance measure.
- ▶ ANOVA relies on r measures.
- ▶ r measures are a function of the correlation between dependent and independent variables.
- ▶ Two measures:
 - ▶ η^2
 - ▶ ω^2

$$\eta^2$$

$$\eta^2 = \frac{SS_{\text{Between}}}{SS_{\text{Total}}}$$

- ▶ Proportion of explained variance (R^2)
- ▶ Percent reduction in error

Example: Fear of Heights

Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Between	2,519.03	4	629.76	5.98
Within	10,006.00	95	105.33	
Total	12,525.03	99		

Example: Fear of Heights

Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Between	2,519.03	4	629.76	5.98
Within	10,006.00	95	105.33	
Total	12,525.03	99		

$$\eta^2 = \frac{2,519.03}{12,525.03} = 0.2011$$

ω^2

η^2 is biased: it overestimates the proportion of “explained” variance.

ω^2 is a correction to η^2 :

$$\omega^2 = \frac{SS_{\text{Between}} - (k - 1)MS_{\text{Within}}}{SS_{\text{Total}} + MS_{\text{Within}}}$$

Example: Fear of Heights

Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Between	2,519.03	4	629.76	5.98
Within	10,006.00	95	105.33	
Total	12,525.03	99		

Example: Fear of Heights

Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Between	2,519.03	4	629.76	5.98
Within	10,006.00	95	105.33	
Total	12,525.03	99		

$$\omega^2 = \frac{2,519.03 - (5 - 1)(105.33)}{12,525.03 + 105.33} = 0.1661$$

ANOVA Effect Size

$$\eta^2 = \frac{SS_{\text{Between}}}{SS_{\text{Total}}}$$

$$\omega^2 = \frac{SS_{\text{Between}} - (k - 1)(MS_{\text{Within}})}{SS_{\text{Total}} + MS_{\text{Within}}}$$

Noncentrality Parameter

$$\begin{aligned} F &\approx \frac{E(MS_{\text{Between}})}{E(MS_{\text{Within}})} \\ &= \frac{\sigma_e^2 + \frac{\sum_{j=1}^k N_j \tau_j^2}{k-1}}{\sigma_e^2} \\ &= 1 + \frac{\sum_{j=1}^k N_j \tau_j^2 / (k-1)}{\sigma_e^2} \end{aligned}$$

Noncentrality Parameter

$$\begin{aligned} F &\approx \frac{E(MS_{\text{Between}})}{E(MS_{\text{Within}})} \\ &= \frac{\sigma_e^2 + \frac{\sum_{j=1}^k N_j \tau_j^2}{k-1}}{\sigma_e^2} \\ &= 1 + \frac{\sum_{j=1}^k N_j \tau_j^2 / (k-1)}{\sigma_e^2} \end{aligned}$$

$$\lambda = \frac{\sum_{j=1}^k N_j \tau_j^2}{\sigma_e^2}$$

$$\hat{\lambda} = \frac{SS_{\text{Between}}}{MS_{\text{Within}}}$$

T-shirt Sizes

For both η^2 and ω^2 ,

Small	0.01
Medium	0.06
Large	0.14

In R

```
> # ANOVA in R
> group.factor <- as.factor(group)
> results <- aov(X~group.factor)
> summary(results)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
group.factor	4	2519	629.8	5.979	0.000246 ***
Residuals	95	10006	105.3		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$$\eta^2 = \frac{2,519.03}{12,525.03} = 0.2011$$

$$\omega^2 = \frac{2,519.03 - (5 - 1)(105.33)}{12,525.03 + 105.33} = 0.1661$$

$$\hat{\lambda} = \frac{2,519.03}{105.33} = 23.9224$$

In R

```
> results <- aov(X~group.factor)
> eta_squared(results)
```

For one-way between subjects designs, partial eta squared is equivalent to etasquared. Returning eta squared.

Effect Size for ANOVA

Parameter	Eta2	95% CI

group.factor	0.20	[0.07, 1.00]

In R

```
> omega_squared(results)
```

For one-way between subjects designs, partial omega squared is equivalent to omega squared. Returning omega squared.

Effect Size for ANOVA

Parameter	Omega2	95% CI

group.factor	0.17	[0.04, 1.00]

More Fear of Heights

ANOVA is regression.

```
> summary.lm(results)
```

Call:

```
aov(formula = X ~ group)
```

Residuals:

Min	1Q	Median	3Q	Max
-22.411	-7.947	1.174	7.009	24.115

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	40.1569	2.2948	17.499	< 2e-16 ***
group2	2.2915	3.2454	0.706	0.482
group3	2.3884	3.2454	0.736	0.464
group4	-0.2687	3.2454	-0.083	0.934
group5	13.3395	3.2454	4.110	8.39e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.26 on 95 degrees of freedom

Multiple R-squared: 0.2011, Adjusted R-squared: 0.1675

F-statistic: 5.979 on 4 and 95 DF, p-value: 0.0002462

See also:

- ▶ Cohen's f

$$f = \sqrt{\frac{\eta^2}{1 - \eta^2}}$$

- ▶ Cohen's f^2

$$f^2 = \frac{\eta^2}{1 - \eta^2}$$

- ▶ Noncentrality parameter

$$\lambda = f^2 \sum_{j=1}^k n_j$$

T-shirt Sizes

Size	η^2, ω^2	f
Small	0.01	0.10
Medium	0.06	0.25
Large	0.14	0.40

Power Analysis in R

```
> # Post-hoc  
> power.f.ancova(eta.squared = 0.2011, factor.levels=5,  
>               k.covariates=0, n.total=100)
```

```
+-----+  
|              POWER CALCULATION              |  
+-----+
```

One-way Analysis of Variance (F-Test)

Hypotheses

H0 (Null Claim) : eta.squared = 0
H1 (Alt. Claim) : eta.squared > 0

Results

Total Sample Size = 100
Type 1 Error (alpha) = 0.050
Type 2 Error (beta) = 0.014
Statistical Power = 0.986 <<

Power Analysis in R

```
> # A priori  
> power.f.ancova(eta.squared = 0.20, factor.levels=5,  
>                k.covariates=0, power=0.80)
```

```
+-----+  
|          SAMPLE SIZE CALCULATION          |  
+-----+
```

One-way Analysis of Variance (F-Test)

Hypotheses

H0 (Null Claim) : eta.squared = 0
H1 (Alt. Claim) : eta.squared > 0

Results

Total Sample Size = 55 <<
Type 1 Error (alpha) = 0.050
Type 2 Error (beta) = 0.179
Statistical Power = 0.821

Post Hoc Tests

Why did we reject $H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$?

Post Hoc Tests

Why did we reject $H_0 : \mu_1 = \mu_2 = \dots = \mu_k$?

- ▶ If the omnibus F is not significant, we stop.
- ▶ If the omnibus F is significant, then we can try and determine why it was significant.

Post Hoc Tests

Why did we reject $H_0 : \mu_1 = \mu_2 = \dots = \mu_k$?

- ▶ If the omnibus F is not significant, we stop.
- ▶ If the omnibus F is significant, then we can try and determine why it was significant.
 - ▶ To make specific comparisons between means, we could perform
 - ▶ Bonferroni or Dunn-Šidák procedure
 - ▶ Newman-Keuls test
 - ▶ Sheffé test
 - ▶ Tukey's HSD test
 - ▶ Duncan's multiple range test

Dunn-Šidák Correction

Remember that

$$\alpha_{FW} = 1 - (1 - \alpha_{PC})^M,$$

where M is the number of comparisons to be made.

Bonferroni Inequality

Remember that

$$\alpha_{FW} = 1 - (1 - \alpha_{PC})^M,$$

where M is the number of comparisons to be made.

$$\alpha_{FW} \leq M\alpha_{PC}$$

Fear of Heights

```
> results <- aov(X ~ group)
> summary(results)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
group.factor	4	2519	629.8	5.979	0.000246 ***
Residuals	95	10006	105.3		

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

> pairwise.t.test(X,group,p.adjust.method = "none")

Pairwise comparisons using t tests with pooled SD

data:  X and group
```

	1	2	3	4
2	0.48186	-	-	-
3	0.46358	0.97624	-	-
4	0.93419	0.43215	0.41498	-
5	8.4e-05	0.00097	0.00107	6.2e-05

```
P value adjustment method: none
```

Fear of Heights

```
> results <- aov(X ~ group)
> summary(results)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
group.factor	4	2519	629.8	5.979	0.000246	***
Residuals	95	10006	105.3			

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

> pairwise.t.test(X,group,p.adjust.method = "bonferroni")

Pairwise comparisons using t tests with pooled SD

data:  X and group
```

	1	2	3	4
2	1.00000	-	-	-
3	1.00000	1.00000	-	-
4	1.00000	1.00000	1.00000	-
5	0.00084	0.00973	0.01073	0.00062

```
P value adjustment method: bonferroni
```

Studentized Range Statistic

Let $\{X_1, X_2, \dots, X_N\}$ be a sample from a $\mathcal{N}(\mu, \sigma)$ distribution. Then

$$q = \frac{\max X - \min X}{s}$$

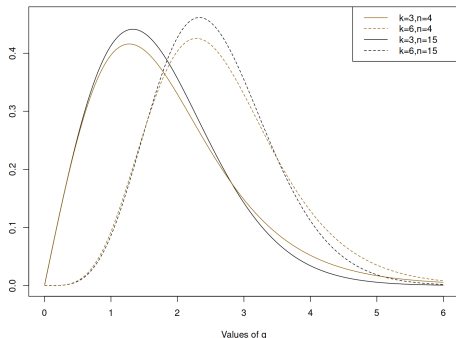
follows the *Studentized range distribution*.

Studentized Range Statistic

Let $\{\bar{X}_1, \bar{X}_2, \dots, \bar{X}_k\}$ be a sample from a $\mathcal{N}(\mu, \sigma/\sqrt{n})$ distribution. Then

$$q = \frac{\bar{X}_{\max} - \bar{X}_{\min}}{s_P/\sqrt{n}}$$

follows the *Studentized range distribution*, where n is the group size.



Tukey's HSD Test

- ▶ Make pairwise comparisons between all means while holding the overall probability of making a Type I error constant.
- ▶ Must determine a critical “honestly significant difference.”
- ▶ Compute the differences between all pairs of means.
- ▶ Any difference that exceeds the critical value is “honestly significant.”

The Honestly Significant Difference

There must be a value of $\bar{X}_{\max} - \bar{X}_{\min}$ that is big enough to reject $H_0 : \mu_1 = \cdots = \mu_k$. This is the Honestly Significant Difference (HSD). Rearrange the expression for q :

$$q_{(\alpha; df, k)} = \frac{HSD}{\sqrt{\frac{MS_W}{n}}}$$

The Honestly Significant Difference

There must be a value of $\bar{X}_{\max} - \bar{X}_{\min}$ that is big enough to reject $H_0 : \mu_1 = \cdots = \mu_k$. This is the Honestly Significant Difference (HSD). Rearrange the expression for q :

$$q_{(\alpha;df,k)} = \frac{HSD}{\sqrt{\frac{MS_W}{n}}}$$
$$HSD = q_{(\alpha;df,k)} \sqrt{\frac{MS_W}{n}}$$

The Honestly Significant Difference

There must be a value of $\bar{X}_{\max} - \bar{X}_{\min}$ that is big enough to reject $H_0 : \mu_1 = \dots = \mu_k$. This is the Honestly Significant Difference (HSD). Rearrange the expression for q :

$$q_{(\alpha; df, k)} = \frac{HSD}{\sqrt{\frac{MS_W}{n}}}$$

$$HSD = q_{(\alpha; df, k)} \sqrt{\frac{MS_W}{n}}$$

MS_W = mean square within from ANOVA

n = number of people in each group

Differences of Means

	40.16	42.45	42.55	39.89	53.5
40.16		-2.29	-2.39	0.27	-13.34
42.45			-0.1	2.56	-11.05
42.55				2.66	-10.95
39.89					-13.61
53.5					

```
> qtuke(.95,5,95) * sqrt(105.3/20)
[1] 9.023895
```

Tukey's HSD Test

```
> TukeyHSD(results,ordered=TRUE)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = X ~ group)

$group
      diff      lwr      upr    p adj
1-4  0.26870729 -8.756314  9.293729 0.9999892
2-4  2.56023271 -6.464789 11.585254 0.9332594
3-4  2.65714727 -6.367874 11.682169 0.9242538
5-4 13.60815945  4.583138 22.633181 0.0005807
2-1  2.29152542 -6.733496 11.316547 0.9546428
3-1  2.38843998 -6.636582 11.413462 0.9475290
5-1 13.33945216  4.314430 22.364474 0.0007841
3-2  0.09691456 -8.928107  9.121936 0.9999998
5-2 11.04792674  2.022905 20.072948 0.0084451
5-3 10.95101218  1.925990 19.976034 0.0092643

> 1 - ptukey(2.65714727 / sqrt(105.3/20),5,95)
[1] 0.9242218
```

What if the groups are of different sizes?

How do we compute n for the HSD?

What if the groups are of different sizes?

How do we compute n for the HSD?

If there are unequal group n_j s, then use

$$n = \frac{k}{1/n_1 + 1/n_2 + \cdots 1/n_k},$$

the *harmonic mean* of the n_j s.

What if the groups are of different sizes?

How do we compute n for the HSD?

Tukey-Kramer approach: Remember the Welch's t -test.

$$s_P \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = \sqrt{\frac{s_P^2}{n_1} + \frac{s_P^2}{n_2}}$$
$$\rightarrow \sqrt{\frac{\frac{MS_{\text{Within}}}{n_i} - \frac{MS_{\text{Within}}}{n_j}}{2}}$$

What if the groups are of different sizes?

How do we compute n for the HSD?

Tukey-Kramer approach: Remember the Welch's t -test.

$$\begin{aligned} s_P \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} &= \sqrt{\frac{s_P^2}{n_1} + \frac{s_P^2}{n_2}} \\ &\rightarrow \sqrt{\frac{\frac{MS_{\text{Within}}}{n_i} - \frac{MS_{\text{Within}}}{n_j}}{2}} \\ &= \sqrt{\frac{MS_{\text{Within}}}{2}} \sqrt{\frac{1}{n_i} + \frac{1}{n_j}} \\ &= \sqrt{MS_{\text{Within}}/n} \quad \text{if } n_i = n_j. \end{aligned}$$

GSS 1980 ANOVA

```
> data <- read.csv("GSS_1980.csv")
> money <- data$REALRINC
> happy <- as.factor(data$HAPPY < 3)
```

```
> gss <- aov(money~happy)
> summary(gss)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
happy	1	3.189e+09	3.189e+09	5.51	0.0191 *
Residuals	871	5.040e+11	5.787e+08		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

GSS 1980 t-test

```
> t.test(money~happy,var.equal=TRUE)
```

Two Sample t-test

```
data: money by happy
t = -2.3473, df = 871, p-value = 0.01913
alternative hypothesis: true difference in means between group FALSE
and group TRUE is not equal to 0
95 percent confidence interval:
 -10877.8729  -970.7736
sample estimates:
mean in group FALSE  mean in group TRUE
      17924.87         23849.19
```

ANOVA vs. t -test

ANOVA ($\alpha = 0.05$)

$$F(1, 871) = 5.51$$

$$F_C(1, 871) = 3.85$$

$$p = 0.019$$

t -test ($\alpha = 0.05$)

$$t(871) = -2.3473$$

$$t_C(871) = \pm 1.9627$$

$$p = 0.0191$$

Therefore,

$$t_{\alpha/2}(df)^2 = F_{\alpha}(1, df) :$$

performing an ANOVA on two groups is mathematically equivalent to performing a *two-tailed* independent samples t test.

Data

Participants rate guilt/innocence on a 10-point scale (1 = certain she is innocent, 10 = certain she is guilty)

Juror	Record	Group		
		Clean	Control	
1	10	5	4	
2	7	1	6	
3	5	3	9	
4	10	7	3	
5	8	4	3	
$\bar{X}_j$	8	4	5	$\bar{\bar{X}} = 5.67$

- ▶ What is the power of this test to detect a medium effect size (η^2 or $\omega^2 = 0.06$)?
- ▶ Compute η^2 and ω^2 .
- ▶ Use the Bonferroni procedure to determine which means differed from the others.
- ▶ Perform Tukey's HSD test to determine which means differed from the others.